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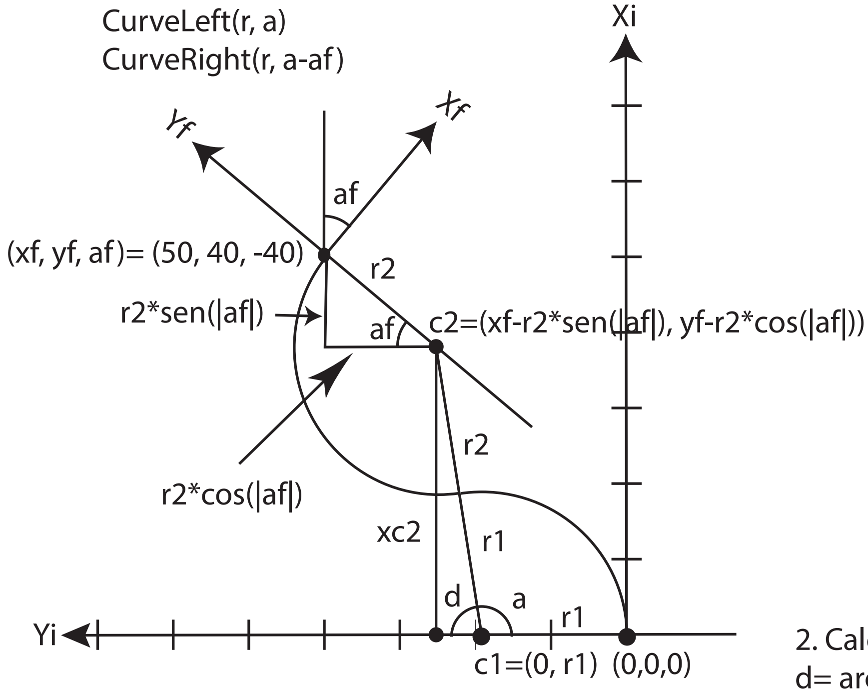
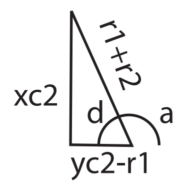
Theoretical trajectory 3

Consider a final point (xf, yf, af).
When xf>=yf and 45>=af>-90 and yc2>yc1,
a possible trajectory is:

CurveLeft(r, a)
CurveRight(r, a-af)





1. Calculate r1 and r2 using distance d12 between points c1 and c2, which define yhe center points of two circunference arcs of each curve:
c1=(xc1, yc1)
xc1= 0 and yc1= r1, then
c1= (0, r1)
and,
c2= (xc2, yc2),
xc2= xf+r2*sen(|af|) and yc2= yf-r2*cos(|af|), then
c2= (xf+r2*sen(|af|), yf-r2*cos(|af|)).

Then,
$d12=((xf+r2*sen(|af|))^2 + (r1-yf+r2*cos(|af|))^2)^{0,5}=r1+r2$.

As the equation d12 has two unknown variables r1 and r2, there are three possible solutions to get only one unknown variable:

i. $r1=r2=r \Rightarrow (2*r)^2= (xf+r*sen(|af|))2 + (r*(cos(|af|)+1)-yf))^2$

ii. $r1= C \Rightarrow (r2+C)^2=(xf+r2*sen(|af|))^2 + (C+r2*cos(|af|)-yf))^2$.

iii. $r2=C \Rightarrow (r1+C)^2=(xf+C*sen(|af|))^2 + (r1+C*cos(|af|)-yf))^2$.

2. Calculate angle a,
d= arcsen((xf-r2*sen(|af|))/(r1+r2))
a= 180-d

3. For example (50,40,-40) and considering equation 1.i for d12, we get
$(2*r)^2=(yf-r*(1+cos(|af|)))^2+(xf-r*sen(|af|))^2$,

calculating the coeficients of resolvent formula,
a=2-2*cos(|af|),
b=2*yf*(1+cos(|af|))+2*xf*sen(|af|),
$c= -(xf^2+yf^2)$

applying the resolvent formula, we get
r=19,1

from equation on point 2,
a=180-arcsen((50-19,1*sen(40))/(2*19,1)
a= 99,4

4. The theoretical trajectory is,
CurveLeft(19.1, 99.4)
CurveRight(19.1, 139.4)

Note: In practice, the robot is not able to do a curve with a radius of 19,1cm due its manobrability constraints in 2D space.